# Progressive block-size sweeps on TU-Berlin sketches

Five families × two inits × two compression rates ($m = n = 8$, $256 \times 256$)

Generated 2026-05-25. Mean test PSNR (dB) over 50 sketches. — = still training; n/a = stage undefined (mera needs $k \in \{2, 4, 8\}$).

## 1 Sketches are a different regime

The TU-Berlin set is line drawings: $\approx 96.5\%$ white pixels with thin black ink. They are **block-sparse** — most $8 \times 8$ tiles are blank — so a classical block transform is near-lossless: block-DCT-8 reaches 100.65 dB at $\rho = 0.2$ (median 94.48), versus $\approx 32$ dB on DIV2K. This inverts the DIV2K story, so we report the learned circuits at **two** compression rates: the headline $\rho = 0.20$ (5×) and an aggressive $\rho = 0.05$ (20×).

The experiment is otherwise identical to the DIV2K matrix: each stage $k$ trains a $2^k \times 2^k$ inner circuit replicated by `BlockedBasis` (bare at $k = 8$), 1008 steps, `MSELoss` top-$k$. Inits: **identity** (gates dropped to identity operator) and **random** (rich Haar $U(2)/U(4)$; qft Haar-on-H + random phases; tebd/entangled_qft/mera native seed). Families run $k = 2..8$ except `mera` ($k \in \{2, 4, 8\}$).

## 2 Light compression: $\rho = 0.20$ (5×)

| family | init | $k$=2 (4) | $k$=3 (8) | $k$=4 (16) | $k$=5 (32) | $k$=6 (64) | $k$=7 (128) | $k$=8 (256) |
|---|---|---|---|---|---|---|---|---|
| rich | identity | 81.1 | 78.8 | 74.9 | 62.1 | 62.1 | 61.5 | 61.5 |
| rich | random | 79.5 | 73.8 | 62.6 | 48.8 | 41.5 | 33.9 | 44.9 |
| qft | identity | 83.6 | 87.8 | 71.4 | 51.6 | 50.1 | 64.2 | 62.6 |
| qft | random | 87.1 | 83.1 | 65.6 | 48.5 | 39.6 | 43.2 | 44.2 |
| tebd | identity | 85 | 87.5 | 69.6 | 51.4 | 50.2 | 62.9 | 62.9 |
| tebd | random | 88.2 | 87.2 | 69.5 | 45.9 | 34.6 | 44.2 | 38.1 |
| ent-qft | identity | 83.6 | 87.8 | 71.4 | 51.6 | 50.1 | 64.2 | 62.6 |
| ent-qft | random | 82.7 | 85.4 | 70.1 | 45.7 | 38.4 | 44.2 | 44.2 |
| mera | identity | 85 | n/a | 69.6 | n/a | n/a | n/a | 62.9 |
| mera | random | 88.2 | n/a | 69.4 | n/a | n/a | n/a | 38.4 |

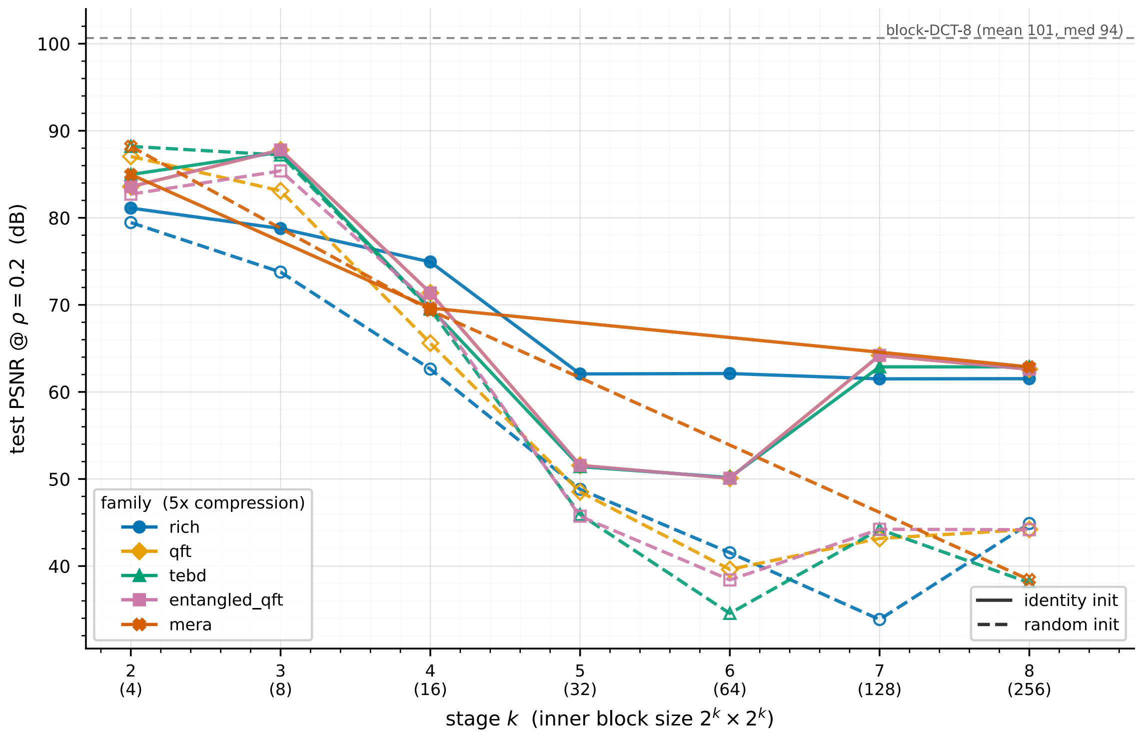


Figure 1: PSNR @ $\rho = 0.20$. Block-sparse sketches make **small** blocks win: the curve falls with $k$. All learned circuits sit below classical block-DCT-8 (grey).

## 3 Heavy compression: $\rho = 0.05$ (20×)

| family | init | $k$=2 (4) | $k$=3 (8) | $k$=4 (16) | $k$=5 (32) | $k$=6 (64) | $k$=7 (128) | $k$=8 (256) |
|---|---|---|---|---|---|---|---|---|
| rich | identity | 7.6 | 37.3 | 37.2 | 34.8 | 35.5 | 35.5 | 36.1 |
| rich | random | 7.6 | 30.6 | 31.6 | 29.6 | 28.5 | 25.7 | 29.3 |
| qft | identity | 7.6 | 33.6 | 31.5 | 29.7 | 29.7 | 31.5 | 31.5 |
| qft | random | 7.6 | 29.8 | 31 | 28.8 | 27.3 | 28 | 28.4 |
| tebd | identity | 7.6 | 33.5 | 31.5 | 29.7 | 29.7 | 31.5 | 31.5 |
| tebd | random | 7.6 | 33.5 | 31.5 | 28.5 | 26.1 | 28.5 | 27.1 |
| ent-qft | identity | 7.6 | 33.6 | 31.5 | 29.7 | 29.7 | 31.5 | 31.5 |
| ent-qft | random | 7.6 | 33.5 | 31.5 | 28.5 | 27.1 | 28.5 | 28.5 |
| mera | identity | 7.6 | n/a | 31.5 | n/a | n/a | n/a | 31.5 |
| mera | random | 7.6 | n/a | 31.5 | n/a | n/a | n/a | 27.1 |

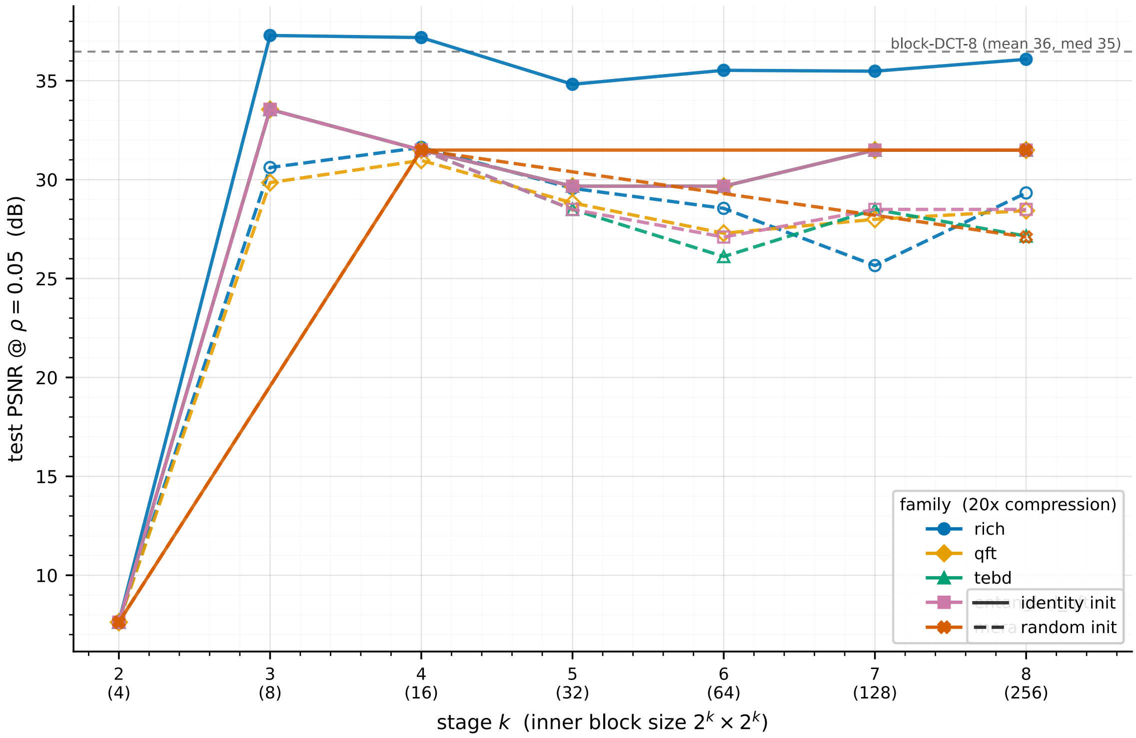


Figure 2: PSNR @ $\rho = 0.05$. At 20 × compression the learned circuits become **competitive with** block-DCT-8 (grey) — rich (identity) edges above it at $k = 3, 4$. The per-$k$ curve also flattens.

## 4 Observations

- **Block sparsity dominates the light-compression regime, and the curve inverts.** At $\rho = 0.20$ PSNR **falls** with block size $k$ — small blocks win because each blank tile costs almost no coefficients — the opposite of DIV2K's flat ≈ 33 dB. The best learned circuits peak at ≈ 88 dB ($k = 3$, the QFT-derived families) but still trail classical block-DCT-8 (median ≈ 94 dB): on near-lossless-compressible sketches the fixed block transform is hard to beat.
- **The family ranking flips between rates.** At $\rho = 0.20$ the QFT-derived families (qft/tebd/entangled_qft) **beat** `rich` at the small blocks (≈ 88 vs 79 dB at $k = 3$). At $\rho = 0.05$ this reverses: `rich` (identity) leads at ≈ 37 dB and even edges **above** block-DCT-8 (≈ 36 dB) at $k = 3, 4$. When only 5% of coefficients survive, learning a genuine energy-compacting transform pays off and the classical near-lossless advantage evaporates.
- **Identity init beats random for every family, by a lot.** On sketches the gap is far larger than on DIV2K — up to ≈ 25–30 dB at large $k$ ($\rho = 0.2$, e.g. rich $k = 7$: 61.5 vs 33.9). A random start is genuinely damaging here.

- **The QFT-family identity-init equivalence partly survives.** Under identity init `qft` and `entangled_qft` are bit-identical at every $k$ (as on DIV2K); at $\rho = 0.05$ all four QFT-derived families coincide, but at $\rho = 0.20$ they split into a qft/entangled_qft pair and a tebd/mera pair. `rich` (complex $U(4)$) stands apart from all of them.

Working artefact (report-first; not yet cellified into `results/`). All combos from `/tmp/tuberlin/<family>_<init>/`; classical refs computed on the same seed=42 / 50-sketch test split.